

Converting X-ray Linear Polarization to Circular Polarization Using a Crystal Wave Plate

Manuel Sanchez del Rio^{1*}, Sonia Francoual², and Carsten Detlefs¹

¹ESRF, Grenoble, France

²DESY, Hamburg, Germany

^{*}*srio@esrf.eu*

July 8, 2026

Abstract

Converting a linearly polarized X-ray beam into a circularly polarized beam is a standard technique at synchrotron radiation facilities. The key optical element is a **quarter-wave plate (QWP)**, a birefringent crystal that introduces a 90° ($\lambda/4$) phase retardation between the two orthogonal components of the electric field.

This document presents a self-contained theoretical treatment covering: (i) the Jones and Stokes formalisms and their connection via the Pauli matrices; (ii) the Jones and Mueller matrices of a general retarder at an arbitrary orientation angle χ ; (iii) the particular case of the QWP converting horizontal linear polarization into right circular polarization; (iv) the experimental Stokes polarimetry protocol used at synchrotron beamlines; and (v) the general crystal phase plate with complex amplitudes r_s , r_p and the resulting degree of circular polarization as a function of χ and the relative phase retardation $\Delta\phi$.

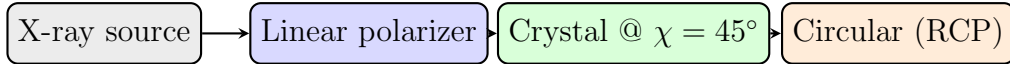
Contents

1	Introduction	3
2	Jones and Mueller–Stokes Calculus Treatment	3
2.1	Jones Vector and Stokes Parameters	3
2.2	Degree of Polarization and Poincaré Sphere	4
2.3	Jones Matrices and Mueller Matrices	5
2.4	Rotation of an Optical Element by Angle χ	5
2.5	Jones and Mueller Matrices of a General Retarder	6
2.6	QWP at $\chi = 45^\circ$: Jones and Mueller Matrices	7
2.7	Input Beam: Horizontal Linear Polarization	7
2.8	Output Beam: Right Circular Polarization	7
2.8.1	Jones route	7
2.8.2	Mueller route	8
2.8.3	Time-domain picture	8
2.8.4	Summary: both routes give RCP	8
2.9	Left Circular Polarization: QWP at $\chi = -45^\circ$	9
2.10	Summary of Key Jones and Mueller Matrices	10
3	Experimental Polarization Analysis	11
3.1	Principle	11
3.2	Transmitted Intensity Formula	11
3.3	Measurements at Key Angles: General Case	11
3.4	Measuring S_3 : Adding a QWP Analyzer	12
3.5	Particular Case: Circular Polarization	13
3.6	Complete Stokes Polarimetry Protocol: General and Circular Case	14
4	Crystal Phase Plate with General Amplitudes r_s, r_p	14
4.1	Physical Context	14
4.2	Jones Matrix of the Crystal	15
4.3	Crystal Oriented at Arbitrary Angle χ	15
4.4	Mueller Matrix of the Crystal at Arbitrary Angle χ	15
4.4.1	Aligned Mueller Matrix ($\chi = 0$)	15
4.4.2	Rotated Mueller Matrix	16
4.5	Output Jones Vector for a General Incident Beam	16
4.6	Output Stokes Vector for a General Incident Beam	17
4.7	Key Result: Degree of Circular Polarization for a General Input	18
4.8	Conditions for Perfect Circular Polarization	20
4.9	Summary	22
5	Experimental benchmarking	24

1 Introduction

Synchrotron X-ray beams produced by bending magnets or undulators are naturally linearly polarized in the horizontal plane of the storage ring. Many X-ray spectroscopy and scattering experiments — magnetic circular dichroism (MCD), resonant magnetic X-ray scattering [1, 2], chiral diffraction — require circularly polarized beams. The standard approach is to insert a **crystal phase plate** in the beam, which acts as the X-ray analogue of a quarter-wave plate [3–5].

To convert horizontal linear polarization to circular polarization, the crystal must be oriented with its birefringence axis at $\chi = 45^\circ$ to the incoming polarization direction, so that both σ - and π -polarization components are equally excited and acquire a relative phase shift of $\pi/2$. The optical layout is:



The optical properties of perfect crystals offer a powerful mechanism for manipulating polarization through birefringence near a Bragg reflection. The σ - and π -polarization components experience different complex reflectivities r_s and r_p , whose relative magnitude and phase can be tuned by rocking the crystal angle. The phase retardation $\Delta\phi = \arg(r_s) - \arg(r_p)$ reaches $\pm\pi/2$ at a detuning of approximately half the Darwin width from the Bragg peak, providing the ideal quarter-wave condition.

Practically, thin (a few hundred microns) perfect crystals (silicon or diamond) are used as phase plates at hard X-ray beamlines (ESRF, APS, SPring-8) because they can be tuned to high near-equal transmission, phase shift close to $\pi/2$, and offer excellent mechanical stability [4, 5].

2 Jones and Mueller–Stokes Calculus Treatment

2.1 Jones Vector and Stokes Parameters

A fully coherent monochromatic beam propagating along $\hat{z}$ is described in the Jones formalism [6] by a two-component complex electric field vector:

$$\mathbf{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} |E_x| e^{i\phi_x} \\ |E_y| e^{i\phi_y} \end{pmatrix} \quad (1)$$

where E_x and E_y are the complex amplitudes of the horizontal ($\hat{x}$) and vertical ($\hat{y}$) field components, and ϕ_x, ϕ_y are their absolute phases.

The four real, observable **Stokes parameters** [7, 8] are defined as:

$$S_0 = |E_x|^2 + |E_y|^2 \quad (2)$$

$$S_1 = |E_x|^2 - |E_y|^2 \quad (3)$$

$$S_2 = 2 \operatorname{Re}(E_x E_y^*) \quad (4)$$

$$S_3 = 2 \operatorname{Im}(E_x E_y^*) \quad (5)$$

S_0 is the total intensity, S_1 measures linear horizontal/vertical polarization preference, S_2 measures $\pm 45^\circ$ linear polarization, and S_3 measures circular polarization content.

Parameter	Measures	Extreme values (normalized)	
S_0	Total intensity	$S_0 > 0$ always	conserved through lossless optics
S_1	H vs. V linear	+1: horiz. (H)	-1: vertical (V)
S_2	$\pm 45^\circ$ linear	+1: $+45^\circ$	-1: -45°
S_3	Circular content	+1: RCP	-1: LCP

Table 1: Physical interpretation of the Stokes parameters (normalized, $S_0 = 1$).

Pauli Matrix Representation

The four **Pauli matrices** (including the 2×2 identity) form a complete orthogonal basis for all 2×2 Hermitian matrices:

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (6)$$

They satisfy the orthogonality relation $\text{Tr}(\sigma_m \sigma_n) = 2\delta_{mn}$ and the product rule $\sigma_j \sigma_k = \delta_{jk} \sigma_0 + i\varepsilon_{jkl} \sigma_l$, where ε_{jkl} is the Levi-Civita symbol.

The **coherency matrix** $\rho = \mathbf{E}\mathbf{E}^\dagger$ provides a 2×2 Hermitian representation of the polarization state:

$$\rho = \begin{pmatrix} \langle E_x E_x^* \rangle & \langle E_x E_y^* \rangle \\ \langle E_y E_x^* \rangle & \langle E_y E_y^* \rangle \end{pmatrix} = \frac{1}{2} \sum_{k=0}^3 S_k \sigma_k \quad (7)$$

The Stokes parameters are then the projections of ρ onto the Pauli basis via the trace inner product:

$$S_k = \text{Tr}(\sigma_k \rho), \quad k = 0, 1, 2, 3 \quad (8)$$

This compact formula reproduces Eqs. (2)–(5) exactly and is the foundation of the Jones-to-Mueller conversion used throughout this document (see Section 2.3).

2.2 Degree of Polarization and Poincaré Sphere

The **degree of polarization** (DOP) quantifies how far a beam departs from complete unpolarized light:

$$\text{DOP} = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0} \in [0, 1] \quad (9)$$

DOP = 1 for fully coherent (polarized) light; DOP = 0 for completely unpolarized light. The degree of linear and circular polarization are:

$$\text{DOLP} = \frac{\sqrt{S_1^2 + S_2^2}}{S_0}, \quad \text{DOCP} = \frac{S_3}{S_0} \in [-1, +1] \quad (10)$$

The **Poincaré sphere** provides a geometric representation: a fully polarized state corresponds to a point on the sphere of radius S_0 at position (S_1, S_2, S_3) .

- **North pole** $(0, 0, +S_3)$: right circular polarization (RCP)
- **South pole** $(0, 0, -S_3)$: left circular polarization (LCP)
- **Equator** $(S_3 = 0)$: all linear polarization states

- $+S_1$ axis: horizontal linear; $-S_1$ axis: vertical linear
- $+S_2$ axis: $+45^\circ$ linear; $-S_2$ axis: -45° linear

Optical elements act as rotations on the Poincaré sphere (for lossless retarders) or as contractions toward the origin (for partial polarizers).

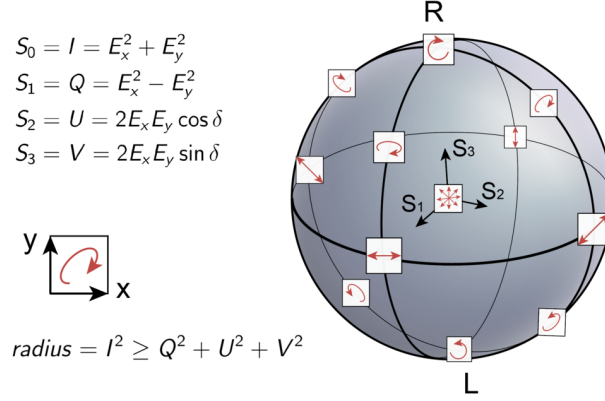


Figure 1: The Poincare sphere. <https://commons.wikimedia.org/w/index.php?curid=90694298>

2.3 Jones Matrices and Mueller Matrices

An optical element transforms the polarization state of a beam. In the two formalisms:

- In the **Jones formalism**, a 2×2 complex matrix $\mathbf{J}$ transforms the Jones vector: $\mathbf{E}' = \mathbf{J} \mathbf{E}$. This is valid for fully coherent beams only.
- In the **Mueller–Stokes formalism** [9, 10], a real 4×4 matrix $\mathbf{M}$ transforms the Stokes vector: $\mathbf{S}' = \mathbf{M} \mathbf{S}$. This is valid for any degree of coherence and polarization.

For a fully coherent beam, the two formalisms are equivalent. The Mueller matrix corresponding to a Jones matrix $\mathbf{J}$ is obtained via the Pauli trace formula. Since the coherency matrix transforms as $\boldsymbol{\rho}' = \mathbf{J} \boldsymbol{\rho} \mathbf{J}^\dagger$, projecting onto the Pauli basis (Eq. 8) gives each Mueller matrix element as:

$$M_{mn} = \frac{1}{2} \text{Tr}(\boldsymbol{\sigma}_m \mathbf{J} \boldsymbol{\sigma}_n \mathbf{J}^\dagger), \quad m, n \in \{0, 1, 2, 3\} \quad (11)$$

When several optical elements are placed in series, their Jones matrices (or Mueller matrices) are multiplied in order from right to left:

$$\mathbf{J}_{\text{total}} = \mathbf{J}_N \cdots \mathbf{J}_2 \mathbf{J}_1, \quad \mathbf{M}_{\text{total}} = \mathbf{M}_N \cdots \mathbf{M}_2 \mathbf{M}_1 \quad (12)$$

2.4 Rotation of an Optical Element by Angle χ

When an optical element is physically rotated by angle χ about the beam axis, its Jones matrix in the lab frame becomes:

$$\mathbf{J}^{(\chi)} = \mathbf{R}(-\chi) \cdot \mathbf{J}^{(0)} \cdot \mathbf{R}(\chi) \quad (13)$$

where the Jones rotation matrix is:

$$\mathbf{R}(\chi) = \begin{pmatrix} \cos \chi & \sin \chi \\ -\sin \chi & \cos \chi \end{pmatrix} \quad (14)$$

The analogous Mueller rotation matrix is:

$$\mathbf{R}_M(\chi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\chi & \sin 2\chi & 0 \\ 0 & -\sin 2\chi & \cos 2\chi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (15)$$

The key property is that a *physical* rotation by χ appears as a rotation by 2χ in Stokes space, because the Stokes parameters are quadratic in the field amplitudes. Note that S_0 (total intensity) and S_3 (circular content) are invariant under rotation, while (S_1, S_2) rotate as a two-dimensional vector.

The rotated Mueller matrix is:

$$\mathbf{M}^{(\chi)} = \mathbf{R}_M(-\chi) \cdot \mathbf{M}^{(0)} \cdot \mathbf{R}_M(\chi) \quad (16)$$

2.5 Jones and Mueller Matrices of a General Retarder

A **linear retarder** (wave plate) introduces a phase retardation δ between its fast and slow eigenpolarization axes. With the fast axis aligned along $\hat{x}$ ($\chi = 0$), the Jones matrix is:

$$\mathbf{J}_{\text{ret}}^{(0)} = e^{-i\delta/2} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix} \quad (17)$$

The global phase factor $e^{-i\delta/2}$ is physically irrelevant for intensity measurements and is often dropped. For a **quarter-wave plate** (QWP), $\delta = \pi/2$:

$$\mathbf{J}_{\text{QWP}}^{(0)} = e^{-i\pi/4} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad (18)$$

Using Eq. (11), the corresponding aligned Mueller matrix is:

$$\mathbf{M}_{\text{QWP}}^{(0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & +1 & 0 \end{pmatrix} \quad (19)$$

This matrix leaves S_0 and S_1 unchanged (H/V polarization unaffected since the fast axis is along $\hat{x}$) and maps $S_2 \rightarrow -S_3$, $S_3 \rightarrow +S_2$: it converts $\pm 45^\circ$ linear polarization into circular polarization.

For a general retarder with fast axis rotated to angle χ , applying Eq. (16) to $\mathbf{M}_{\text{ret}}^{(0)}$ gives the full Mueller matrix:

$$\mathbf{M}_{\text{ret}}^{(\chi, \delta)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\chi + \sin^2 2\chi \cos \delta & \sin 2\chi \cos 2\chi (1 - \cos \delta) & +\sin 2\chi \sin \delta \\ 0 & \sin 2\chi \cos 2\chi (1 - \cos \delta) & \sin^2 2\chi + \cos^2 2\chi \cos \delta & -\cos 2\chi \sin \delta \\ 0 & -\sin 2\chi \sin \delta & +\cos 2\chi \sin \delta & \cos \delta \end{pmatrix} \quad (20)$$

One verifies immediately: $\delta = \pi/2$, $\chi = 0$ recovers Eq. (19) ✓.

2.6 QWP at $\chi = 45^\circ$: Jones and Mueller Matrices

Applying the rotation sandwich (Eq. 13) to $\mathbf{J}_{\text{QWP}}^{(0)}$ with $\chi = 45^\circ$:

$$\mathbf{J}_{\text{QWP}}^{(45^\circ)} = \mathbf{R}(-45^\circ) \cdot e^{-i\pi/4} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \cdot \mathbf{R}(45^\circ) = \frac{e^{-i\pi/4}}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad (21)$$

Setting $\delta = \pi/2$ and $\chi = 45^\circ$ in Eq. (20):

$$\mathbf{M}_{\text{QWP}}^{(45^\circ)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \quad (22)$$

This matrix swaps $S_1 \leftrightarrow -S_3$ while leaving S_0 and S_2 unchanged. It converts horizontal linear polarization ($S_1 = 1, S_3 = 0$) into right circular polarization ($S_1 = 0, S_3 = 1$).

Equivalently, applying $\mathbf{R}_M(-45^\circ) \mathbf{M}_{\text{QWP}}^{(0)} \mathbf{R}_M(45^\circ)$ with $\mathbf{R}_M(45^\circ)$ evaluated at $2\chi = 90^\circ$ yields the same result as Eq. (22). $\checkmark$

2.7 Input Beam: Horizontal Linear Polarization

The Jones vector of a horizontally polarized beam (normalized, $E_0 = 1$) is:

$$\mathbf{E}_{\text{in}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (23)$$

Applying Eqs. (2)–(5):

$$\mathbf{S}_{\text{in}} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (24)$$

This state lies on the equator of the Poincaré sphere at the $+S_1$ pole.

2.8 Output Beam: Right Circular Polarization

2.8.1 Jones route

Applying Eq. (21) to the input Jones vector:

$$\mathbf{E}_{\text{out}} = \mathbf{J}_{\text{QWP}}^{(45^\circ)} \mathbf{E}_{\text{in}} = \frac{e^{-i\pi/4}}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{e^{-i\pi/4}}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (25)$$

Dropping the global phase $e^{-i\pi/4}$:

$$\mathbf{E}_{\text{out}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (26)$$

This is the canonical Jones vector for right circular polarization. Computing the Stokes parameters from Eqs. (2)–(5):

$$S_0 = \left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2} + \frac{1}{2} = 1 \quad (27)$$

$$S_1 = \left| \frac{1}{\sqrt{2}} \right|^2 - \left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2} - \frac{1}{2} = 0 \quad (28)$$

$$S_2 = 2 \operatorname{Re} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} \right) = 2 \operatorname{Re} \left(\frac{-i}{2} \right) = 0 \quad (29)$$

$$S_3 = 2 \operatorname{Im} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} \right) = 2 \operatorname{Im} \left(\frac{-i}{2} \right) = +1 \quad (30)$$

2.8.2 Mueller route

Applying $\mathbf{M}_{\text{QWP}}^{(45^\circ)}$ (Eq. 22) to $\mathbf{S}_{\text{in}}$ (Eq. 24):

$$\mathbf{S}_{\text{out}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \quad (31)$$

The sign of S_3 depends on the adopted convention for the sense of circular polarization. With the convention of Eqs. (5) and (26), $S_3 = +1$ corresponds to RCP. The Mueller matrix $\mathbf{M}_{\text{QWP}}^{(45^\circ)}$ as written here uses the opposite sign convention (IEEE/optics convention where $S_3 > 0$ is LCP), giving $S_3 = -1$ for the same physical RCP state. Both are self-consistent within their own conventions.

2.8.3 Time-domain picture

In real space and time the output field components are:

$$E_x(t) = \frac{1}{\sqrt{2}} \cos \omega t, \quad E_y(t) = \frac{1}{\sqrt{2}} \sin \omega t \quad (32)$$

The tip of the electric field vector traces a **circle** of radius $1/\sqrt{2}$ in the (x, y) plane, confirming circular polarization. The rotation is counter-clockwise when viewed looking toward the source (RCP by the physics convention adopted here).

2.8.4 Summary: both routes give RCP

$$\mathbf{S}_{\text{out}} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ +1 \end{pmatrix} \quad (\text{RCP}), \quad \text{DOCP} = +1 \quad (33)$$

The state sits at the **north pole** of the Poincaré sphere.

2.9 Left Circular Polarization: QWP at $\chi = -45^\circ$

Rotating the QWP to $\chi = -45^\circ$ reverses the sign of the off-diagonal terms in the Jones matrix:

$$\mathbf{J}_{\text{QWP}}^{(-45^\circ)} = \frac{e^{-i\pi/4}}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \implies \mathbf{E}_{\text{out}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad \mathbf{S}_{\text{out}} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \quad (\text{LCP}) \quad (34)$$

The corresponding Mueller matrix is:

$$\mathbf{M}_{\text{QWP}}^{(-45^\circ)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (35)$$

which maps horizontal linear ($S_1 = 1$) to $S_3 = -1$ (LCP), as expected.

2.10 Summary of Key Jones and Mueller Matrices

Element	Jones matrix J	Mueller matrix M
QWP, fast axis at $\chi = 0^\circ$	$e^{-i\pi/4} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & +1 & 0 \end{pmatrix}$
QWP, fast axis at $\chi = +45^\circ$	$\frac{e^{-i\pi/4}}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$
QWP, fast axis at $\chi = -45^\circ$	$\frac{e^{-i\pi/4}}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$
General retarder		
fast axis at χ , retardation δ	$e^{-i\delta/2} \mathbf{R}(-\chi) \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix} \mathbf{R}(\chi)$ see Eq. (20)	
Rotation by χ	$\begin{pmatrix} \cos \chi & \sin \chi \\ -\sin \chi & \cos \chi \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\chi & \sin 2\chi & 0 \\ 0 & -\sin 2\chi & \cos 2\chi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Table 2: Jones and Mueller matrices for key optical elements. The doubling $\chi \rightarrow 2\chi$ in the Mueller rotation matrix reflects the quadratic dependence of Stokes parameters on the field amplitudes.

3 Experimental Polarization Analysis

3.1 Principle

A linear polarizer placed in the beam transmits only the projection of the electric field onto its transmission axis. By recording the transmitted intensity $I(\theta)$ as a function of the polarizer rotation angle θ , and by optionally inserting a quarter-wave plate (QWP) analyzer before the polarizer, all four Stokes parameters of an *arbitrary* polarization state (S_0, S_1, S_2, S_3) can be determined.

3.2 Transmitted Intensity Formula

For a linear polarizer at angle θ to the horizontal, the Jones matrix is:

$$\mathbf{J}_{LP}(\theta) = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix} \quad (36)$$

The Mueller matrix of a linear polarizer at angle θ is:

$$\mathbf{M}_{LP}(\theta) = \frac{1}{2} \begin{pmatrix} 1 & \cos 2\theta & \sin 2\theta & 0 \\ \cos 2\theta & \cos^2 2\theta & \sin 2\theta \cos 2\theta & 0 \\ \sin 2\theta & \sin 2\theta \cos 2\theta & \sin^2 2\theta & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (37)$$

Applying this to an arbitrary incident Stokes vector $\mathbf{S} = (S_0, S_1, S_2, S_3)^T$, the detected intensity is the first component of $\mathbf{M}_{LP}(\theta) \mathbf{S}$:

$$I(\theta) = \frac{1}{2} (S_0 + S_1 \cos 2\theta + S_2 \sin 2\theta) \quad (38)$$

This is a **sinusoidal function of 2θ** with mean $S_0/2$, cosine amplitude $S_1/2$, and sine amplitude $S_2/2$. Note that S_3 does *not* appear: a linear polarizer alone is insensitive to circular polarization.

3.3 Measurements at Key Angles: General Case

Four strategically chosen orientations of the polarizer are sufficient to determine S_0 , S_1 , and S_2 from Eq. (38).

$\theta = 0^\circ$ (**Horizontal transmission axis**)

$$I_H \equiv I(0^\circ) = \frac{1}{2} (S_0 + S_1) \quad (39)$$

$\theta = 90^\circ$ (**Vertical transmission axis**)

$$I_V \equiv I(90^\circ) = \frac{1}{2} (S_0 - S_1) \quad (40)$$

$\theta = 45^\circ$ (**+45° transmission axis**)

$$I_{45} \equiv I(45^\circ) = \frac{1}{2} (S_0 + S_2) \quad (41)$$

$\theta = 135^\circ$ (-45° **transmission axis**)

$$I_{135} \equiv I(135^\circ) = \frac{1}{2}(S_0 - S_2) \quad (42)$$

From these four measurements the first three Stokes parameters are recovered by simple linear combinations:

$$S_0 = I_H + I_V = I_{45} + I_{135} \quad (43)$$

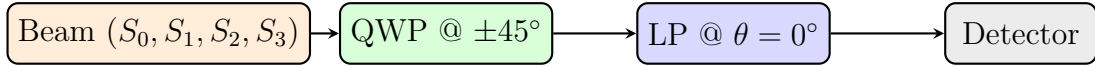
$$S_1 = I_H - I_V \quad (44)$$

$$S_2 = I_{45} - I_{135} \quad (45)$$

Equations (43)–(45) are valid for *any* incident polarization state. They show that S_1 is the contrast between horizontal and vertical transmissions, and S_2 is the contrast between the two diagonal transmissions.

3.4 Measuring S_3 : Adding a QWP Analyzer

Since a linear polarizer alone cannot detect S_3 , a QWP must be inserted immediately before the polarizer. Its role is to convert circular polarization into linear polarization that the polarizer can then analyse.



Applying $\mathbf{M}_{\text{QWP}}^{(\pm 45^\circ)}$ (Eq. 22) to the general Stokes vector, then transmitting through LP at $\theta = 0^\circ$:

$$I_+ \equiv I[\text{QWP}(+45^\circ), \text{LP}(0^\circ)] = \frac{1}{2}(S_0 + S_3) \quad (46)$$

$$I_- \equiv I[\text{QWP}(-45^\circ), \text{LP}(0^\circ)] = \frac{1}{2}(S_0 - S_3) \quad (47)$$

The general relation for S_3 is:

$$S_3 = I_+ - I_- = \frac{I_{\text{QWP}(+45^\circ)} - I_{\text{QWP}(-45^\circ)}}{1} \quad (48)$$

or equivalently, normalized to the total intensity:

$$\text{DOCP} = \frac{S_3}{S_0} = \frac{I_+ - I_-}{I_+ + I_-} \quad (49)$$

Configuration	Detected intensity	Stokes relation
LP at $\theta = 0^\circ$	$I_H = \frac{1}{2}(S_0 + S_1)$	$S_1 = 2I_H - S_0$
LP at $\theta = 90^\circ$	$I_V = \frac{1}{2}(S_0 - S_1)$	$S_0 = I_H + I_V$
LP at $\theta = 45^\circ$	$I_{45} = \frac{1}{2}(S_0 + S_2)$	$S_2 = 2I_{45} - S_0$
LP at $\theta = 135^\circ$	$I_{135} = \frac{1}{2}(S_0 - S_2)$	check: $S_0 = I_{45} + I_{135}$
QWP(+45°)+LP(0°)	$I_+ = \frac{1}{2}(S_0 + S_3)$	$S_3 = I_+ - I_-$
QWP(-45°)+LP(0°)	$I_- = \frac{1}{2}(S_0 - S_3)$	$\text{DOCP} = (I_+ - I_-)/(I_+ + I_-)$

Table 3: General six-measurement Stokes polarimetry protocol, valid for any incident polarization state (S_0, S_1, S_2, S_3) .

3.5 Particular Case: Circular Polarization

For the RCP beam produced by the crystal QWP, the Stokes vector is $\mathbf{S} = I_0(1, 0, 0, +1)^T$, i.e. $S_1 = S_2 = 0$ and $S_3 = S_0 = I_0$. Substituting into the general formulas:

Linear polarizer scan.

$$I(\theta) = \frac{1}{2}(I_0 + 0 \cdot \cos 2\theta + 0 \cdot \sin 2\theta) = \frac{I_0}{2} \quad (50)$$

$$I(\theta) = \frac{I_0}{2} \quad \forall \theta \quad (51)$$

The intensity is **constant regardless of polarizer angle** — the definitive fingerprint of a fully circularly polarized beam. Because $S_1 = S_2 = 0$, the sinusoidal modulation of Eq. (38) vanishes completely.

Key angle measurements. From Eqs. (39)–(42):

$$I_H = I_V = I_{45} = I_{135} = \frac{I_0}{2} \quad (52)$$

All four linear polarizer positions give the same reading, confirming $S_1 = S_2 = 0$.

QWP analyzer for S_3 . From Eqs. (46)–(47):

$$I_+ = \frac{1}{2}(I_0 + I_0) = I_0 \quad (\text{maximum}) \quad (53)$$

$$I_- = \frac{1}{2}(I_0 - I_0) = 0 \quad (\text{minimum}) \quad (54)$$

$$S_3 = I_+ - I_- = I_0 - 0 = I_0, \quad \text{DOCP} = \frac{I_+ - I_-}{I_+ + I_-} = \frac{I_0}{I_0} = +1 \quad \checkmark \quad (55)$$

3.6 Complete Stokes Polarimetry Protocol: General and Circular Case

Meas.	Configuration	General result	Stokes relation	RCP value
I_0	No optics	S_0	$S_0 = I_0$	I_0
I_H	LP at $\theta = 0^\circ$	$\frac{1}{2}(S_0 + S_1)$	$S_1 = 2I_H - S_0$	$I_0/2$
I_V	LP at $\theta = 90^\circ$	$\frac{1}{2}(S_0 - S_1)$	$S_0 = I_H + I_V$	$I_0/2$
I_{45}	LP at $\theta = 45^\circ$	$\frac{1}{2}(S_0 + S_2)$	$S_2 = 2I_{45} - S_0$	$I_0/2$
I_{135}	LP at $\theta = 135^\circ$	$\frac{1}{2}(S_0 - S_2)$	check: $I_{45} + I_{135} = S_0$	$I_0/2$
I_+	QWP(+45°)+LP(0°)	$\frac{1}{2}(S_0 + S_3)$	$S_3 = I_+ - I_-$	I_0
I_-	QWP(-45°)+LP(0°)	$\frac{1}{2}(S_0 - S_3)$	DOCP = $(I_+ - I_-)/(I_+ + I_-)$	0

Table 4: Complete Stokes polarimetry protocol. The general result column applies to any incident polarization state. The RCP value column gives the numerical outcome for the beam $\mathbf{S} = I_0(1, 0, 0, 1)^T$ produced by the crystal QWP. θ is the polarizer transmission-axis angle measured from horizontal.

This six-measurement (or seven, counting I_0) protocol is the standard **Stokes polarimetry** procedure used at X-ray synchrotron beamlines. In practice S_0 need not be measured separately since $S_0 = I_H + I_V = I_{45} + I_{135} = I_+ + I_-$, providing three independent consistency checks.

4 Crystal Phase Plate with General Amplitudes r_s, r_p

4.1 Physical Context

In the X-ray regime, a birefringent crystal phase plate exploits the difference in reflectivity (and phase shift) between the σ (s) and π (p) polarization components near a Bragg reflection [3, 11]. The two eigenpolarizations of the crystal are:

- **σ -polarization** (s): electric field perpendicular to the diffraction plane — mapped to the x -axis.
- **π -polarization** (p): electric field in the diffraction plane — mapped to the y -axis.

The complex reflection (or transmission) amplitudes r_s and r_p are generally different in both magnitude and phase:

$$r_s = |r_s| e^{i\phi_s}, \quad r_p = |r_p| e^{i\phi_p} \quad (56)$$

The key quantity is the **relative phase retardation**:

$$\Delta\phi = \phi_p - \phi_s = \arg(r_p) - \arg(r_s) \quad (57)$$

For a perfect quarter-wave plate, $r_s r_s^* = r_p r_p^*$ and $\arg(r_p/r_s) = \pm\pi/2$. All Mueller matrix entries and Stokes output formulas below are expressed directly in terms of the complex amplitudes r_s , r_p and their conjugates, using the standard products $r_s r_s^* = |r_s|^2$, $r_p r_p^* = |r_p|^2$, and $r_s r_p^* = \text{Re}(r_s r_p^*) + i \text{Im}(r_s r_p^*)$.

4.2 Jones Matrix of the Crystal

With the crystal axes aligned along $\hat{x}$ (σ) and $\hat{y}$ (π), the Jones matrix is diagonal:

$$\mathbf{J}_0 = \begin{pmatrix} r_s & 0 \\ 0 & r_p \end{pmatrix} \quad (58)$$

Using the Pauli basis (Eq. 6), this decomposes as:

$$\mathbf{J}_0 = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} \boldsymbol{\sigma}_1 \quad (59)$$

This is physically transparent: the crystal acts as an isotropic attenuator (the $\boldsymbol{\sigma}_0$ term, proportional to the mean amplitude) plus a linear H/V birefringence (the $\boldsymbol{\sigma}_1$ term, proportional to the amplitude difference).

4.3 Crystal Oriented at Arbitrary Angle χ

Applying the rotation sandwich $\mathbf{J}^{(\chi)} = \mathbf{R}(-\chi) \mathbf{J}_0 \mathbf{R}(\chi)$ and using the SU(2) identity $\mathbf{R}(-\chi) \boldsymbol{\sigma}_1 \mathbf{R}(\chi) = \cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2$, the rotated Pauli decomposition is:

$$\mathbf{J}^{(\chi)} = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} (\cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2) \quad (60)$$

Rotation by χ simply rotates the birefringence axis from $\boldsymbol{\sigma}_1$ (H/V) toward $\boldsymbol{\sigma}_2$ ($\pm 45^\circ$) by the angle 2χ in Pauli/Stokes space. Expanded in matrix form:

$$\mathbf{J}^{(\chi)} = \frac{r_s + r_p}{2} \mathbf{I} + \frac{r_s - r_p}{2} \begin{pmatrix} \cos 2\chi & -\sin 2\chi \\ -\sin 2\chi & -\cos 2\chi \end{pmatrix} \quad (61)$$

4.4 Mueller Matrix of the Crystal at Arbitrary Angle χ

4.4.1 Aligned Mueller Matrix ($\chi = 0$)

Applying the Pauli trace formula (Eq. 11) to $\mathbf{J}_0$:

$$\mathbf{M}_0 = \begin{pmatrix} \frac{r_s r_s^* + r_p r_p^*}{2} & \frac{r_s r_s^* - r_p r_p^*}{2} & 0 & 0 \\ \frac{r_s r_s^* - r_p r_p^*}{2} & \frac{r_s r_s^* + r_p r_p^*}{2} & 0 & 0 \\ 0 & 0 & \text{Re}(r_s r_p^*) & \text{Im}(r_s r_p^*) \\ 0 & 0 & -\text{Im}(r_s r_p^*) & \text{Re}(r_s r_p^*) \end{pmatrix} \quad (62)$$

The block structure is transparent: the upper-left block is driven by $r_s r_s^*$ and $r_p r_p^*$ (the individual intensity reflectivities), while the lower-right 2×2 block is the real 2×2 rotation matrix $\text{Re}(r_s r_p^*) \mathbf{I} + \text{Im}(r_s r_p^*) \boldsymbol{\epsilon}$ (where $\boldsymbol{\epsilon}$ is the 2×2 antisymmetric matrix), encoding the complex cross-term $r_s r_p^*$ between the two eigenmodes.

4.4.2 Rotated Mueller Matrix

Using the Mueller rotation matrix $\mathbf{R}_M(\chi)$ (Eq. 15):

$$\mathbf{M}^{(\chi)} = \mathbf{R}_M(-\chi) \cdot \mathbf{M}_0 \cdot \mathbf{R}_M(\chi) \quad (63)$$

Unlike the Jones case, the rotation couples all four Stokes components. Carrying out the full matrix product with $c_2 \equiv \cos 2\chi$, $s_2 \equiv \sin 2\chi$:

$$\mathbf{M}^{(\chi)} = \begin{pmatrix} \frac{r_s r_s^* + r_p r_p^*}{2} & \frac{r_s r_s^* - r_p r_p^*}{2} c_2 & \frac{r_s r_s^* - r_p r_p^*}{2} s_2 & 0 \\ \frac{r_s r_s^* - r_p r_p^*}{2} c_2 & \frac{r_s r_s^* + r_p r_p^*}{2} c_2^2 + \text{Re}(r_s r_p^*) s_2^2 & \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 & -\text{Im}(r_s r_p^*) s_2 \\ \frac{r_s r_s^* - r_p r_p^*}{2} s_2 & \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 & \frac{r_s r_s^* + r_p r_p^*}{2} s_2^2 + \text{Re}(r_s r_p^*) c_2^2 & \text{Im}(r_s r_p^*) c_2 \\ 0 & \text{Im}(r_s r_p^*) s_2 & -\text{Im}(r_s r_p^*) c_2 & \text{Re}(r_s r_p^*) \end{pmatrix} \quad (64)$$

Key structural features:

- $M_{00} = (r_s r_s^* + r_p r_p^*)/2$ and $M_{33} = \text{Re}(r_s r_p^*)$ are independent of χ .
- The last row $(0, \text{Im}(r_s r_p^*) s_2, -\text{Im}(r_s r_p^*) c_2, \text{Re}(r_s r_p^*))$ shows that S_3^{out} is driven by S_1^{in} and S_2^{in} (via $\text{Im}(r_s r_p^*)$) and by S_3^{in} (via $\text{Re}(r_s r_p^*)$).
- When $r_s r_s^* = r_p r_p^*$ (equal reflectivities) the upper-left block becomes $(r_s r_s^*/2 + r_p r_p^*/2) \mathbf{I}_{2 \times 2}$, and the off-diagonal entries in the first two rows/columns vanish.

4.5 Output Jones Vector for a General Incident Beam

[remove this section? duplicated? trivial?]

Let the incident beam be an arbitrary Jones vector:

$$\mathbf{E}_{\text{in}} = \begin{pmatrix} E_x^{(0)} \\ E_y^{(0)} \end{pmatrix} \quad (65)$$

Applying the crystal Jones matrix (Eq. 61):

$$\mathbf{E}_{\text{out}} = \frac{r_s + r_p}{2} \begin{pmatrix} E_x^{(0)} \\ E_y^{(0)} \end{pmatrix} + \frac{r_s - r_p}{2} \begin{pmatrix} \cos 2\chi E_x^{(0)} - \sin 2\chi E_y^{(0)} \\ -\sin 2\chi E_x^{(0)} - \cos 2\chi E_y^{(0)} \end{pmatrix} \quad (66)$$

The first term is an isotropic multiplication by the mean amplitude $(r_s + r_p)/2$; the second introduces differential retardation along the rotated eigenmodes. Equivalently, using Eq. (60) the output can be written in the Pauli basis as:

$$\mathbf{E}_{\text{out}} = \left[\frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} (\cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2) \right] \mathbf{E}_{\text{in}} \quad (67)$$

Particular case: horizontal linear polarization. Setting $E_x^{(0)} = E_0$, $E_y^{(0)} = 0$ in Eq. (66):

$$\mathbf{E}_{\text{out}}|_H = \frac{E_0}{2} \begin{pmatrix} (r_s + r_p) + (r_s - r_p) \cos 2\chi \\ -(r_s - r_p) \sin 2\chi \end{pmatrix} = \frac{E_0}{2} \begin{pmatrix} (r_s + r_p) + (r_s - r_p) \cos 2\chi \\ (r_p - r_s) \sin 2\chi \end{pmatrix} \quad (68)$$

Particular case: vertical linear polarization. Setting $E_x^{(0)} = 0$, $E_y^{(0)} = E_0$:

$$\mathbf{E}_{\text{out}}|_V = \frac{E_0}{2} \begin{pmatrix} -(r_s - r_p) \sin 2\chi \\ (r_s + r_p) - (r_s - r_p) \cos 2\chi \end{pmatrix} = \frac{E_0}{2} \begin{pmatrix} (r_p - r_s) \sin 2\chi \\ (r_s + r_p) + (r_p - r_s) \cos 2\chi \end{pmatrix} \quad (69)$$

4.6 Output Stokes Vector for a General Incident Beam

Let the incident beam have an arbitrary Stokes vector $\mathbf{S}_{\text{in}} = (S_0^{\text{in}}, S_1^{\text{in}}, S_2^{\text{in}}, S_3^{\text{in}})^T$. Applying the rotated Mueller matrix $\mathbf{M}^{(\chi)}$ (Eq. 64):

$$\mathbf{S}_{\text{out}} = \mathbf{M}^{(\chi)} \mathbf{S}_{\text{in}} \quad (70)$$

Computing each component from Eq. (64), with $c_2 = \cos 2\chi$, $s_2 = \sin 2\chi$:

$$S_0^{\text{out}} = \frac{r_s r_s^* + r_p r_p^*}{2} S_0^{\text{in}} + \frac{r_s r_s^* - r_p r_p^*}{2} c_2 S_1^{\text{in}} + \frac{r_s r_s^* - r_p r_p^*}{2} s_2 S_2^{\text{in}} \quad (71)$$

$$S_1^{\text{out}} = \frac{r_s r_s^* - r_p r_p^*}{2} c_2 S_0^{\text{in}} + \left(\frac{r_s r_s^* + r_p r_p^*}{2} c_2^2 + \text{Re}(r_s r_p^*) s_2^2 \right) S_1^{\text{in}} \\ + \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 S_2^{\text{in}} - \text{Im}(r_s r_p^*) s_2 S_3^{\text{in}} \quad (72)$$

$$S_2^{\text{out}} = \frac{r_s r_s^* - r_p r_p^*}{2} s_2 S_0^{\text{in}} + \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 S_1^{\text{in}} \\ + \left(\frac{r_s r_s^* + r_p r_p^*}{2} s_2^2 + \text{Re}(r_s r_p^*) c_2^2 \right) S_2^{\text{in}} + \text{Im}(r_s r_p^*) c_2 S_3^{\text{in}} \quad (73)$$

$$S_3^{\text{out}} = \text{Im}(r_s r_p^*) s_2 S_1^{\text{in}} - \text{Im}(r_s r_p^*) c_2 S_2^{\text{in}} + \text{Re}(r_s r_p^*) S_3^{\text{in}} \quad (74)$$

Key observations:

- S_3^{out} is driven by $\text{Im}(r_s r_p^*)$ (coupling from S_1^{in} and S_2^{in}) and by $\text{Re}(r_s r_p^*)$ (preservation of S_3^{in}).
- **H/V linear content (S_1^{in}) generates circular polarization** via $\text{Im}(r_s r_p^*) \sin 2\chi$, provided $\chi \neq 0^\circ, 90^\circ$ and $\text{Im}(r_s r_p^*) \neq 0$.
- S_0^{out} is independent of S_3^{in} .

Particular case: horizontal linear polarization input. $\mathbf{S}_{\text{in}} = I_0(1, 1, 0, 0)^T$. Substituting into Eqs. (71)–(74):

$$S_0^{\text{out}} = \frac{I_0}{2} (r_s r_s^* + r_p r_p^* + (r_s r_s^* - r_p r_p^*) \cos 2\chi) = I_0 (r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi) \quad (75)$$

$$S_1^{\text{out}} = \frac{I_0}{2} (r_s r_s^* - r_p r_p^*) \cos 2\chi + I_0 \left(\frac{r_s r_s^* + r_p r_p^*}{2} \cos^2 2\chi + \text{Re}(r_s r_p^*) \sin^2 2\chi \right) \quad (76)$$

$$S_2^{\text{out}} = I_0 \sin 2\chi \left[\frac{r_s r_s^* - r_p r_p^*}{2} + \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) \cos 2\chi \right] \quad (77)$$

$$S_3^{\text{out}} = I_0 \text{Im}(r_s r_p^*) \sin 2\chi \quad (78)$$

S_3^{out} (Eq. 78) is non-zero and comes from $M_{31} = \text{Im}(r_s r_p^*) \sin 2\chi$ acting on $S_1^{\text{in}} = I_0$. The transmitted intensity is $S_0^{\text{out}} = I_0(r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi)$ (Eq. 75). For a QWP ($r_s r_s^* = r_p r_p^* = r^2$, $\text{Re}(r_s r_p^*) = 0$) at $\chi = 45^\circ$, Eqs. (75)–(78) give:

$$\mathbf{S}_{\text{out}} = I_0 r^2 \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \quad (\text{pure circular, } |P_C| = 1) \quad \checkmark \quad (79)$$

(The sign of S_3 reflects the handedness convention.)

For clarity, the three-step rotation-sandwich makes the mechanism explicit. Step 1: $\mathbf{R}_M(\chi) \mathbf{S}_{\text{in}} = I_0(1, \cos 2\chi, -\sin 2\chi, 0)^T$. The $S_2^{\text{crystal}} = -I_0 \sin 2\chi$ component feeds into the $M_0^{23} = \text{Im}(r_s r_p^*)$ entry in step 2, yielding $S_3^{\text{crystal}} = I_0 \text{Im}(r_s r_p^*) \sin 2\chi$, which is invariant in step 3.

Particular case: $+45^\circ$ linear polarization input. $\mathbf{S}_{\text{in}} = I_0(1, 0, 1, 0)^T$:

$$S_0^{\text{out}} = \frac{I_0}{2} (r_s r_s^* + r_p r_p^* + (r_s r_s^* - r_p r_p^*) \sin 2\chi) \quad (80)$$

$$S_1^{\text{out}} = I_0 \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) \sin 2\chi \cos 2\chi \quad (81)$$

$$S_2^{\text{out}} = I_0 \left(\frac{r_s r_s^* + r_p r_p^*}{2} \sin^2 2\chi + \text{Re}(r_s r_p^*) \cos^2 2\chi \right) \quad (82)$$

$$S_3^{\text{out}} = -I_0 \text{Im}(r_s r_p^*) \cos 2\chi \quad (83)$$

For a QWP at $\chi = 0^\circ$ ($\text{Re}(r_s r_p^*) = 0$, $r_s r_s^* = r_p r_p^* = r^2$): $S_3^{\text{out}} = -I_0 \text{Im}(r_s r_p^*)$. At $\chi = 45^\circ$: $S_3^{\text{out}} = 0$.

Particular case: right circular polarization input. $\mathbf{S}_{\text{in}} = I_0(1, 0, 0, 1)^T$:

$$S_0^{\text{out}} = \frac{I_0}{2} (r_s r_s^* + r_p r_p^*) \quad (84)$$

$$S_1^{\text{out}} = -I_0 \text{Im}(r_s r_p^*) \sin 2\chi \quad (85)$$

$$S_2^{\text{out}} = I_0 \text{Im}(r_s r_p^*) \cos 2\chi \quad (86)$$

$$S_3^{\text{out}} = I_0 \text{Re}(r_s r_p^*) \quad (87)$$

For a QWP ($\text{Re}(r_s r_p^*) = 0$): $S_3^{\text{out}} = 0$, confirming conversion to linear polarization for any χ .

These results are consistent with the output Jones vector of Eq. (66) (up to the sign convention for S_3 , which depends on the choice of circular polarization handedness).

4.7 Key Result: Degree of Circular Polarization for a General Input

Using Eqs. (74) and (71), the degree of circular polarization for an arbitrary incident beam is:

$$P_C(\chi, \mathbf{S}_{\text{in}}) = \frac{\text{Im}(r_s r_p^*) (s_2 S_1^{\text{in}} - c_2 S_2^{\text{in}}) + \text{Re}(r_s r_p^*) S_3^{\text{in}}}{\frac{r_s r_s^* + r_p r_p^*}{2} S_0^{\text{in}} + \frac{r_s r_s^* - r_p r_p^*}{2} (c_2 S_1^{\text{in}} + s_2 S_2^{\text{in}})} \quad (88)$$

with $c_2 = \cos 2\chi$, $s_2 = \sin 2\chi$. Key observations:

1. $\text{Im}(r_s r_p^*)$ **drives the generation of circular polarization** from S_1^{in} and S_2^{in} .
2. $\text{Re}(r_s r_p^*)$ **preserves existing circular polarization** S_3^{in} — it equals zero for an ideal QWP.
3. **H/V linear content** (S_1^{in}) **generates circular polarization** via $\text{Im}(r_s r_p^*) \sin 2\chi$, provided $\chi \neq 0^\circ, 90^\circ$.
4. S_0^{in} **does not contribute to** S_3^{out} .

Particular case: horizontal linear polarization input. For $\mathbf{S}_{\text{in}} = I_0(1, 1, 0, 0)^T$, using Eqs. (75) and (78):

$$P_C(\chi)|_H = \frac{\text{Im}(r_s r_p^*) \sin 2\chi}{r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi} \quad (89)$$

This is the equation in Giles' thesis and in [5], which is written as

$$P_C(\chi)|_H = -\frac{a \sin(\phi_s - \phi_p) \sin 2\chi}{\cos^2 \chi + a^2 \sin^2 \chi}$$

where $a = |r_p|/|r_s|$

For a QWP ($r_s r_s^* = r_p r_p^* = r^2$, $r_p = i r_s$ so $\text{Im}(r_s r_p^*) = -r^2$) at $\chi = 45^\circ$: $P_C = -1$. ✓

Particular case: +45° linear polarization input. For $\mathbf{S}_{\text{in}} = I_0(1, 0, 1, 0)^T$:

$$P_C(\chi)|_{+45} = \frac{-\text{Im}(r_s r_p^*) \cos 2\chi}{\frac{r_s r_s^* + r_p r_p^*}{2} + \frac{r_s r_s^* - r_p r_p^*}{2} \sin 2\chi} \quad (90)$$

For equal reflectivities ($r_s r_s^* = r_p r_p^* = r^2$): $P_C = -\text{Im}(r_s r_p^*) \cos 2\chi / r^2$.

Particular case: right circular polarization input. For $\mathbf{S}_{\text{in}} = I_0(1, 0, 0, 1)^T$:

$$P_C(\chi)|_{\text{RCP}} = \frac{2 \text{Re}(r_s r_p^*)}{r_s r_s^* + r_p r_p^*} \quad (91)$$

For a QWP ($\text{Re}(r_s r_p^*) = 0$): $P_C = 0$ — confirming conversion to linear polarization for any χ .

Input $\mathbf{S}_{\text{in}}$		General P_C (any r_s, r_p)	Ideal QWP ($r_s r_s^* = r_p r_p^* = r^2$, $\text{Re}(r_s r_p^*) = 0$)
H (1, 1, 0, 0) ^T	linear	$\frac{\text{Im}(r_s r_p^*) \sin 2\chi}{r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi}$	$ P_C = 1$ at $\chi = 45^\circ$
+45° (1, 0, 1, 0) ^T	linear	$\frac{-\text{Im}(r_s r_p^*) \cos 2\chi}{\frac{r_s r_s^* + r_p r_p^*}{2} + \frac{r_s r_s^* - r_p r_p^*}{2} \sin 2\chi}$	$P_C = -\frac{\text{Im}(r_s r_p^*)}{r^2} \cos 2\chi$
RCP (1, 0, 0, 1) ^T		$\frac{2 \text{Re}(r_s r_p^*)}{r_s r_s^* + r_p r_p^*}$	$P_C = 0, \forall \chi$
General (S_0, S_1, S_2, S_3) ^T		see Eq. (88)	

Table 5: Degree of circular polarization $P_C = S_3^{\text{out}}/S_0^{\text{out}}$ in terms of the complex amplitudes r_s and r_p . The cross-product $r_s r_p^*$ fully encodes the retardation: $\text{Im}(r_s r_p^*)$ drives conversion between linear and circular polarization, while $\text{Re}(r_s r_p^*)$ governs preservation of existing circular polarization.

4.8 Conditions for Perfect Circular Polarization

For a horizontal linear input (Eq. 89), $|P_C| = 1$ requires $S_1^{\text{out}} = S_2^{\text{out}} = 0$ simultaneously (Eqs. 76–77), which leads to two independent conditions:

1. **Phase condition:** $\text{Re}(r_s r_p^*) = 0$, i.e. $r_s r_p^*$ is purely imaginary (crystal acts as a quarter-wave retarder).
2. **Amplitude-angle condition:** $r_s r_s^* \cos \chi = r_p r_p^* \sin \chi$, i.e. $\tan \chi = \sqrt{r_s r_s^* / r_p r_p^*}$.

For equal reflectivities $|r_s| = |r_p|$, condition 2 reduces to $\chi = 45^\circ$. For unequal reflectivities, the crystal must be tilted to compensate.

For a +45° linear input, condition 1 alone ($\Delta\phi = \pm\pi/2$) is sufficient for $|P_C| = 1$ at $\chi = 0^\circ$. For a general input $(S_0, S_1, S_2, S_3)^T$, perfect circular polarization is achieved whenever the numerator of Eq. (88) equals $\pm S_0^{\text{out}}$.

Input	χ	$\Delta\phi$	$ r_s $ vs $ r_p $	P_C	Output
H linear	0° or 90°	any	any	0	Linear (no conversion)
	45°	$\pm\pi/2$	$ r_s = r_p $	± 1	Pure RCP/LCP
	45°	$\pm\pi/2$	$ r_s \neq r_p $	$ P_C < 1$	Elliptical
	45°	0 or π	any	0	Linear
	$\arctan(r_s / r_p)$	$\pm\pi/2$	any	± 1	Pure RCP/LCP
+45° linear	any	$\pm\pi/2$	any	∓ 1	Pure LCP/RCP
+45° linear	any	0	any	0	Linear (unchanged)
RCP	any	0	any	+1	RCP (unchanged)
RCP	any	$\pi/2$	$ r_s = r_p $	0	+45° linear

Table 6: Output polarization state for selected input polarizations, crystal orientations and parameters. The H linear results use Eq. (89); others use the general Eq. (88).

4.9 Summary

Stage / case	Jones vector E	Pauli of J	decomp.	Stokes matrix	vector / Mueller
Input					
(horiz. linear)	$E_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$	—		$\mathbf{S}_{\text{in}} = I_0$	$\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$
Ideal QWP					
$(\chi = 45^\circ,$					
$ r_s = r_p = r,$					
$\Delta\phi = \pi/2)$	$\frac{E_0 r}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$		$\mathbf{J}_{\text{QWP}} = \frac{r e^{-i\pi/4}(1+i)}{2} \boldsymbol{\sigma}_0 + \frac{r e^{-i\pi/4}(1-i)}{2} \boldsymbol{\sigma}_1$	$= \mathbf{S}_{\text{out}} = I_0 r^2$	$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \text{ (RCP)}$
					$\mathbf{M}_{\text{QWP}}^{(45^\circ)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$

Table 7: Unified summary. $\Delta\phi = \phi_p - \phi_s$ is the relative phase retardation between the π and σ eigenmodes.

Stage / case	Jones vector E	Pauli of J	decomp.	Stokes vector / Mueller matrix
Crystal @ $\chi = 0$				
$(r_s, r_p \text{ general})$	$E_0 \begin{pmatrix} r_s \\ 0 \end{pmatrix}$	$\mathbf{J}_0 = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} \boldsymbol{\sigma}_1$	$\mathbf{M}_0 = \begin{pmatrix} A & B & 0 & 0 \\ B & A & 0 & 0 \\ 0 & 0 & C & D \\ 0 & 0 & -D & C \end{pmatrix}$	$P_C = 0$ (no circular polarization)
Crystal @ arb. χ ,				
$r_s, r_p \text{ general}$	$E_0 \begin{pmatrix} r_s \cos^2 \chi + r_p \sin^2 \chi \\ (r_p - r_s) \sin \chi \cos \chi \end{pmatrix}$	$\mathbf{J}^{(\chi)} = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_p}{2} (\cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2)$	$\mathbf{M}^{(\chi)}$: see Eq. (64)	$S_0 = I_0(r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi)$ $S_3 = I_0 \text{Im}(r_s r_p^*) \sin 2\chi$ P_C : see Eq. (89)

Table 8: Continuation of Table 7

5 Experimental benchmarking

We consider a Si 111 plate of $10\mu\text{m}$ thickness.

The incident beam is linear polarized in the horizontal plane $\mathbf{S}_{\text{in}} = (1, 1, 0, 0)^T$. This is an approximations of the beam emitted by a plane undulator.

The crystal plate is set to $\chi = 45^\circ$. This allows to split the total intensity initially in the H component, into two axis at $\pm 45^\circ$ each one receiving half intensity.

The scan is in the deviation angle from Bragg position. The r_s and r_p calculated correspond to the Bragg geometry, i.e. the angle between the Bragg planes and the crystal surface (zero as it is symmetric) is smaller than the Bragg angle. We consider the transmitted beam (also called forward diffracted).

Experimental data recorded at DESY P9 beamline is in Fig. 5. P3 is not obtained experimentally, but calculated: $P3 = \sqrt{1 - P1^2 - P2^2}$. Note that the sign of the square root is arbitrary (it cannot distinguish between left or right polarization). We selected the positive one.

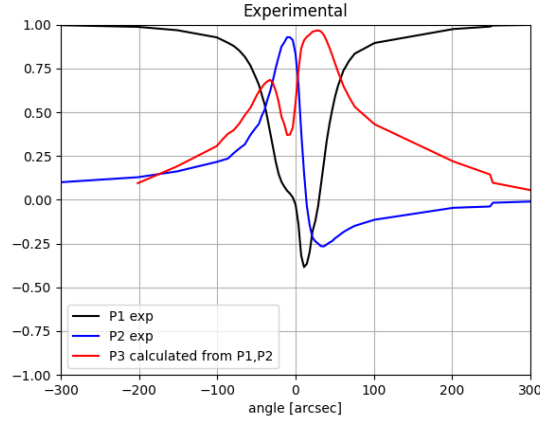


Figure 2: Experimental results.

Calculations using equation 64 are shown in figure 5. The reflectivities r_s and r_p are calculated using the `crystalpy` Python library [12]. We display P1, P2 and P3 (the Stokes parameters normalized to S0). S3 is the circular polarization degree. For direct comparison, the deviation angle in the calculations has been reversed (definitions used to be checked).

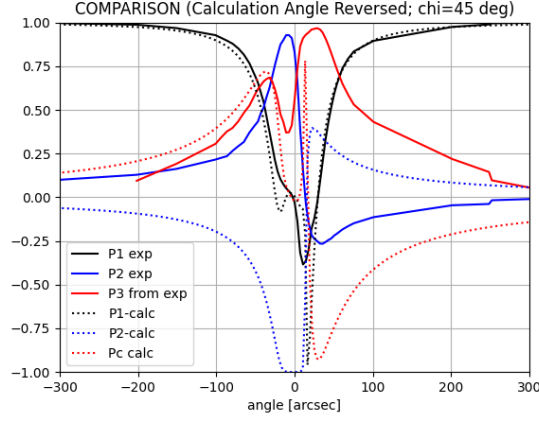


Figure 3: Calculations for $\chi = \pi/4$ and experimental results.

We can observe good agreement in P1, but in P2 and P3 the sign is changed.

For $\chi = -\pi/4$, the results are in figure 5, showing now a good agreement in P1 and P2 (not in P3).

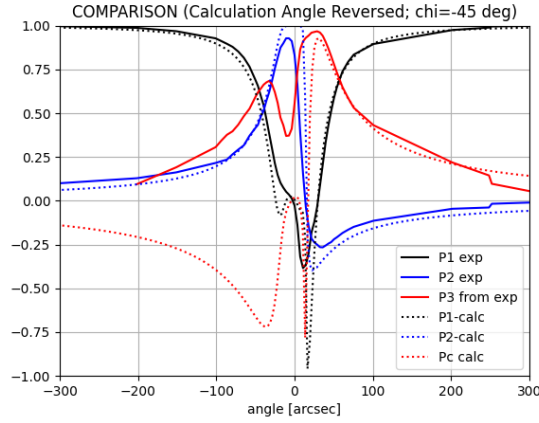


Figure 4: Calculations for $\chi = -\pi/4$ and experimental results.

Note: calculated data are not convoluted. **To be added.**

References

- [1] M. Blume and Doon Gibbs. Polarization dependence of magnetic x-ray scattering. *Physical Review B*, 37(4):1779–1789, 1988. doi: 10.1103/PhysRevB.37.1779.
- [2] M. Blume. Magnetic scattering of X-rays. *Journal of Applied Physics*, 57(8):3615–3618, 1985. doi: 10.1063/1.335023.
- [3] C. Detlefs, M. Sanchez del Rio, and C. Mazzoli. X-ray polarization: General formalism and polarization analysis. *The European Physical Journal Special Topics*, 208(1):359–371, 2012. doi: 10.1140/epjst/e2012-01630-3.

- [4] K. Hirano, K. Izumi, T. Ishikawa, S. Annaka, and S. Kikuta. An X-ray phase plate using Bragg-case diffraction. *Japanese Journal of Applied Physics*, 30(3B):L407–L410, 1991. doi: 10.1143/JJAP.30.L407.
- [5] C. Giles, C. Malgrange, J. Goulon, F. de Bergevin, C. Vettier, E. Dartyge, A. Fontaine, C. Giorgetti, and S. Pizzini. Energy-dispersive phase plate for magnetic circular dichroism experiments in the x-ray range. *Journal of Applied Crystallography*, 27(3):232–240, 1994.
- [6] R. Clark Jones. A new calculus for the treatment of optical systems. I. description and discussion of the calculus. *Journal of the Optical Society of America*, 31(7):488–493, 1941. doi: 10.1364/JOSA.31.000488.
- [7] Max Born and Emil Wolf. *Principles of Optics*. Cambridge University Press, 7th (expanded) edition, 2002. ISBN 978-0-521-64222-4.
- [8] Edward Collett. *Polarized Light: Fundamentals and Applications*. Marcel Dekker, New York, 1993.
- [9] Hans Mueller. The foundation of optics. *Journal of the Optical Society of America*, 38:661, 1948. Abstract only; full lecture presented at the Spring Meeting of the Optical Society of America, 1943.
- [10] Russell A. Chipman, Wai-Sze T. Lam, and Garam Young. Polarized light, fundamentals and applications. In *Handbook of Optics*, volume I, chapter 15. McGraw-Hill, 3rd edition, 2018.
- [11] V. A. Belyakov and V. E. Dmitrienko. Polarisation phenomena in X-ray optics. *Soviet Physics Uspekhi*, 32(8):697–719, 1989.
- [12] Jean-Pierre Guigay and Manuel Sanchez del Rio. Formulation of perfect-crystal diffraction from takagi–taupin equations: numerical implementation in the crystalpy library. *Journal of Synchrotron Radiation*, 31(6):1469–1480, October 2024. ISSN 1600-5775. doi: 10.1107/s160057752400924x. URL <http://dx.doi.org/10.1107/S160057752400924X>.